



NATIONAL OCCUPATIONAL STANDARD FOR A PRECISION AGRICULTURAL TECHNOLOGIST

**NOS.PAT.01
FIRST EDITION**

APPROVING AUTHORITY

This National Occupational Standard has been prepared and published under the authority of the Zambia Qualifications Authority Board on 19th May 2022.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards shall be revised after every **5 years**, or whenever necessary, by the issue of either amendments or of revised editions. It is important that users of National Occupational Standards (NOS) ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Agriculture National Occupational Standards Development Team, upon which the following organisations were represented:

1. Agricultural Institution of Zambia (AIZ)
2. Aquaculture Development Association of Zambia
3. CropLife Zambia
4. Ministry of Fisheries and Livestock (Department of Fisheries)
5. Golden Valley Agricultural Research Trust (GART)
6. Katete College of Agricultural Marketing
7. Ministry of Agriculture (Department of Agriculture)
8. Mulungushi University (MU)
9. Natural Resources Development College (NRDC)
10. University of Zambia (UNZA)
11. Veterinary Council of Zambia
12. Zambia Agriculture Research Institute (ZARI)
13. Zambia National Farmers Union (ZNFU)
14. Zambia Seed Trade Association (ZASTA)
15. Zambia Qualifications Authority (ZAQA)rs Union (ZNFU)

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FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to “**provide for the development and implementation of a national qualifications framework; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing**”.

Among other functions, ZAQA is responsible for “**determining national standards for any occupation**”, through various sector specific National Occupational Standards Development Teams (NOSDTs) of experts composed of representation from appropriate authorities, government departments, industry, academia, regulators, consumer associations and non-governmental organizations, etc.

This National Occupational Standard (NOS) has been developed by the Agriculture National Occupational Standards Development Team in accordance with the procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula and learning programmes in various sectors where the occupation exists. In the Agriculture sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

JUSTIFICATION

Agricultural mechanization is an imperative for the growth of the agriculture sector in Zambia against the backdrop of a declining and static contribution to the country's Gross Domestic Product. The declining agricultural productivity is attributed to, among other things, poor technologies and limited well-trained manpower in the agricultural engineering occupation. Although there is personnel at the level of Agricultural Mechanic, Agricultural Machinery Technologist and Agricultural Engineer with special skills to handle all the technical aspects of the machinery, there is need for specialized personnel to work with Precision agriculture technology that is increasingly seen as a new frontier of sustainable agriculture. Precision agriculture is a crop and livestock production management system that uses a global positioning system (GPS) to monitor equipment field position to collect information and apply inputs as required at each location. Precision agriculture technologies include equipment guidance and automatic steering, yield monitoring, variable rate input application, remote sensing, in-field electronic sensors, section and row control on planters, sprayers and fertilizer applicators, and spatial data management systems. Precision agriculture (PA) involves collecting continuous information about the conditions of agricultural production, and using predictive analytics software to furnish farmers with decision support about the inter- and intra-variability in cropping and livestock systems. These technologies are deployed with the goal of ensuring profitability, sustainability and environmental protection. Because Precision agriculture depends upon specialized equipment, software and IT services, there is a highly recognised need for Precision Agricultural Technologists, who will provide technical services in the utilization of precision agriculture technology in Zambia.

ACRONYMS AND ABBREVIATIONS

CS	Core Skill
EIZ	Engineering Institution of Zambia
GIS	Geographical Information System
GPS	Global Positioning System
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organizational Knowledge
PAM	Precision Agricultural Technologist
PC	Performance Criteria
PS	Professional Skill
RK	Regulatory Knowledge
RPL	Recognition of Prior Learning
TK	Technical Knowledge
ZAQA	Zambia Qualifications Authority
ZICTA	Zambia Information Communications and Technology Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Function: is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles, which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

1. OVERVIEW

The National Occupational Standard for a Precision Agricultural Technologist is premised on the need for specialised personnel to provide technical services to the practice of Precision Agriculture in Zambia as profiled in the table below.

NOS Code	NOS. PAT.01
Occupation	Precision Agricultural Technologist
Job Title	Precision Agricultural Technologist
Job Description	A Precision Agricultural Technologist provides technical services of setting up, operating, repairing and maintaining precision agricultural machines, equipment, tools, sensors and decision-support systems for context-specific management of agricultural production operations and management processes thereby ensuring efficiency, profitability and environmental sustainability.
Job Purpose	To ensure that precision agricultural technology is functional and reliable to achieve the desired level of efficiency for optimal agricultural production and management.
ZQF Level	6
Sector	Agriculture
Sub sector	Agricultural Engineering
Other Economic Sector(s) in which the Occupation is Practiced	Education, Water, Lands, Forestry, Wildlife, Environment, Mining, <i>etc.</i>
Other Similar Jobs that can be performed by a Precision Agricultural Technologist	Geospatial Technologist Systems Analyst Sales Engineering Technologist
Minimum Educational Job Entry Qualification(s)	Diploma
Practicing License Requirements (if any)	Engineering Institution of Zambia Agricultural Institution of Zambia
Training/RPL	Precision Agricultural Mechanics Mechatronics
Minimum Job Entry Age	21 years
Prior Experience	Evidence of three (3) years of vocational training in precision agriculture or evidence of a minimum of six (6) months of internship/apprenticeship in precision agriculture.
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This National Occupational Standard specifies the fundamental knowledge and understanding, skills and competencies that a Precision Agricultural Technologist must possess to be successful in his/her job role. It is applicable to individuals working in public or private organisations or in self-employment.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

This job requires an individual to possess:

- Creativity
- Problem solving skills
- Analytical skills
- Mathematical skills
- Integrity and respect for confidentiality
- Interpersonal skills
- Commercial Awareness
- Attention to details
- Ability to communicate effectively and clearly
- Self-motivated and team worker
- Ability to plan and prioritize,
- Quality consciousness
- Occupational health and safety oriented

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into 6 units representing the tasks that a job holder should undertake in his/her day to day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This unit is about Functional Design of a Precision Agricultural Production Management System]

Unit No.	01
Unit Title	Functional Design of a Precision Agricultural Production Management System
Description	This unit profiles the functional knowledge, skills and competencies as well as job accountabilities needed for the jobholder to effectively set up a Precision Agricultural Production Management System.
Scope	This unit covers the following: • Setting up a Precision Agricultural Production Management System • Calibration of Precision agriculture sensors and devices
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Setting up a Precision Agricultural Production Management System	To be competent, the individual must be able to: PC1. Identify the precision farming requirements for a farm including software needs and applications. PC2. Select appropriate technology for precision agriculture technology including Automatic tractor guidance systems, Variable rate technology for chemical input applicators, surveying equipment, specialised computer mapping software, etc.
Calibration of Precision agriculture sensors and devices	To be competent, the individual must be able to: PC3. Program precision agriculture equipment, tools, sensors and devices for operation. PC4. Calibrate sensors as well as electronic and mechanical control devices for precision agricultural operations. PC5. Calibrate input applications with machinery attachments that can vary the rate of application from GPS controls.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Legislation, regulations, policies, standards, and procedures followed in the company relevant to own employment and performance conditions OK2. Organisational culture, vision and mission OK3. Typical customer profile OK4. Company's service level agreements and policies OK5. Company's code of conduct OK6. Occupational health and safety policies OK7. Environmental management plans

B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. Crop production and management systems - TK2. Animal production and management systems - TK3. Precision farming cycle and components - TK4. Geospatial technologies including Global positioning systems (GPS) and Geographical Information Systems (GIS) - TK5. Electronics and process control - TK6. Electronics and instrumentation - TK7. Computer software programming - TK8. Computational modelling and simulation - TK9. Fundamentals of mechatronics - TK10. Engineering principles of agricultural machines - TK11. Agricultural mechanics - TK12. Agricultural machinery management techniques and methods - TK13. Project management techniques
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Regulatory framework in the sector
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions.
	Reading Skills
	The individual on the job must be able to: - CS2. Read and interpret internal/external documents. - CS3. Read and understand manuals, health and safety instructions, memos and other company documents. - CS4. Read from different sources- books, screens in machines and signage. - CS5. Interpret the various colour codes, nomenclature and acronyms related to the profession.
B. Professional Skills	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : - CS6. Express statements or information clearly so that others can hear and understand. - CS7. Participate in and understand the main points of simple discussions. - CS8. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to:

	PS1. Follow organization rule-based decision-making process.
	PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to:
	PS3. Plan and organise work to meet deadlines.
	PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to:
	PS5. Follow code of conduct.
	PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to:
	PS7. Recognize problems and search for solutions.
	PS8. Choose best methods to complete assigned tasks.
	PS9. Approach relevant authority when required.
	PS10. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to:
	PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to:
	PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities.
	PS13. Ask relevant questions for better understanding.

UNIT 2 [This unit is about operating and managing precision agriculture technology for crop production]

Unit No.	02
Unit Title	Operating and Managing Precision Agriculture Technology for Crop Production
Description	This unit profiles the functional knowledge, skills and competencies as well as job accountabilities needed for the job holder to operate and manage Precision Agriculture Technology for crop production.
Scope	This unit covers the following: • Precision land preparation • Precision planting • Precision nutrient management • Precision water management • Precision crop protection • Precision yield monitoring and mapping
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Precision land preparation	To be competent, the individual must be able to: PC1. Calibrate machinery, equipment and sensors for land development and levelling based on geo-referenced, 3D topographic data. PC2. Generate prescription data for site-specific configuration of the land for a given crop. PC3. Operate a tractor with a GPS-guided tillage implement for variable tillage of depth.
Precision planting	To be competent, the individual must be able to: PC4. Calibrate precision planting equipment for accurate placement of seed at the desired depth and spacing.
Precision nutrient management	To be competent, the individual must be able to: PC5. Use data loggers or soil sensors to collect geo-referenced data on soil attributes of interest including soil pH, organic matter, texture, structure, nutrients, moisture, etc. PC6. Co-ordinate with an authorised soil science laboratory to get the soil samples tested, ensuring the sample is collected from the field, packed and labelled as per the applicable procedure. PC7. Monitor the variability in the soil attributes using variable rate technology. PC8. Generate site-specific maps and reports of the spatial variability of soil attributes. PC9. Use information on the spatial variability of soil attributes to make decisions on the micro and macro-nutrient requirements of the soil as well as the rate, timing and placement of nutrients. PC10. Generate nutrient response functions for crops to be used to prepare soil nutrition supplementation calendars

	PC11. Make site-specific fertiliser recommendations to ensure optimum crop production.
Precision water management	To be competent, the individual must be able to: PC12. Determine the crop water requirements for a specific cropping pattern. PC13. Set up an appropriate irrigation system such as surface irrigation, drip irrigation, or sub-surface irrigation system based on the specified cropping pattern. PC14. Set up a network of sensors for monitoring the performance of various components of an irrigation system. PC15. Delineate irrigation management zones based on spatial sampling with sensor technologies. PC16. Assess the spatio-temporal variability of moisture and plant water use in response to crop physiology, uncertainties in climatic inputs, soil, irrigation systems and water supply limitations, economic considerations and the quality of sensor feedback. PC17. Generate irrigation schedules and strategies such as deficit or full irrigation based on the spatio-temporal variability of moisture and plant water use.
Precision crop protection	To be competent, the individual must be able to: PC18. Plan crop pest monitoring activities. PC19. Assess the spatial and temporal dynamics of weeds, pests and diseases. PC20. Schedule crop protection interventions. PC21. Select appropriate pesticides based on multiple and combined factors such as the target organisms, developmental stage of growth, pesticide movement on plant surfaces and/or inside the plant tissues, etc PC22. Apply crop protection products with precision equipment, including integrated electronic commutation, variable rate technology and drones. PC23. Recommend cropping plans and crop rotation schemes.
Precision yield monitoring and mapping	To be competent, the individual must be able to: PC24. Calibrate a Yield monitor to provide accurate yield and moisture data. PC25. Operate a harvesting machine installed with a Yield monitor. PC26. Process yield maps in a GIS environment. PC27. Analyse yield variability to devise site-specific strategies for improving productivity and profitability.
Knowledge and Understanding (K)	

A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: - OK1. Legislation, regulations, policies, standards, and procedures followed in the company relevant to own employment and performance conditions - OK2. Organisational culture, vision and mission - OK3. Typical customer profile - OK4. Company's service level agreements and policies - OK5. Company's code of conduct - OK6. Occupational health and safety policies - OK7. Environmental management plans
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. Principles and practices of crop production and management - TK2. Design and management of irrigation and drainage systems - TK3. Farm power and machinery - TK4. Precision farming cycle and components - TK5. Agricultural machinery management techniques and methods - TK6. Geospatial technologies including Global positioning systems (GPS) and Geographical Information Systems (GIS) - TK7. Electronics and process control - TK8. Electronics and instrumentation - TK9. Computer software programming - TK10. Computational modelling and simulation - TK11. System optimisation
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Regulatory framework in the sector

A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions.
	Reading Skills
	The individual on the job must be able to: - CS2. Read and interpret internal/external documents. - CS3. Read and understand manuals, health and safety instructions, memos and other company documents. - CS4. Read from different sources- books, screens in machines and signage. - CS5. Interpret the various colour codes, nomenclature and acronyms related to the profession.
Oral Communication (Listening and Speaking skills)	

	The individual on the job must be able to : CS6. Express statements or information clearly so that others can hear and understand. CS7. Participate in and understand the main points of simple discussions. CS8. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities. PS13. Ask relevant questions for better understanding.

UNIT 3 [This unit is about operating and managing precision agriculture technology for livestock production]

Unit No.	03
Unit Title	Operating and Managing Precision Agriculture Technology for Livestock Production
Description	This unit profiles the functional knowledge, skills and competencies as well as job accountabilities needed for the job holder to operate and manage Precision agriculture technology for livestock production.
Scope	This unit covers the following: • Precision electronic identification of animals • Precision environmental control of livestock buildings • Precision animal feeding • Precision monitoring of animal health and welfare • Precision animal product harvesting
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Precision electronic identification of animals	To be competent, the individual must be able to: PC1. Insert a microchip, or electronic transponder, embedded in a tag, bolus or implant for electronic identification of an animal. PC2. Calibrate sensors, devices and tools for electronic identification of animals. PC3. Use electronic devices and aerial vehicles for tracing individual animals.
Precision environmental control of livestock buildings	To be competent, the individual must be able to: PC4. Install sensors and probes for monitoring temperature, light, moisture, air quality and movement, dust, odours, and disease agents. PC5. Maintain the livestock building environment within the comfort zones of the animals using automated electronic controls. PC6. Evaluate the performance of environmental control systems.
Precision animal feeding	To be competent, the individual must be able to: PC7. Use feed formulation software to determine the optimal feed requirements for individual animals or groups of animals. PC8. Calibrate automated feeding systems (AFS) for cost-effective management of animal feeding regimes. PC9. Determine the weight of individual or groups of animals with acceptable precision by correlating feeding to weight gain.
Precision monitoring of animal health and welfare	To be competent, the individual must be able to: PC10. Monitor animal health and welfare parameters in a continuous and automated way, offering the opportunity to improve productivity and detect health issues at an early stage.

	PC11. Record numerous attributes of an individual animal such as pedigree, age, reproduction, growth, health, feed conversion, killing out percentage (carcass weight as percentage of its live weight) and meat quality. PC12. Using signal analysis algorithms and statistical analysis to assess animal grazing behaviour, infection, aggression, weight, feed and water intake, etc.
Precision animal product harvesting	To be competent, the individual must be able to: PC13. Calibrate sensors and tools on animal product harvesting machines such as milking parlours, egg collectors, etc. PC14. Operate animal product harvesting machines and equipment.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Legislation, regulations, policies, standards, and procedures followed in the company relevant to own employment and performance conditions OK2. Organisational culture, vision and mission OK3. Typical customer profile OK4. Company's service level agreements and policies OK5. Company's code of conduct OK6. Occupational health and safety policies OK7. Environmental management plans
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Principles and practices of animal production and management TK2. Farm power and machinery TK3. Precision farming cycle and components TK4. Agricultural machinery management techniques and methods TK5. Geospatial technologies including Global positioning systems (GPS) and Geographical Information Systems (GIS) TK6. Electronics and process control TK7. Electronics and instrumentation TK8. Computer software programming TK9. Computational modelling and simulation TK10. System optimisation
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of rules and regulations of the: RK1. Regulatory framework applicable in the sector

Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: Write in English and give simple concise instructions.
	Reading Skills
	The individual on the job must be able to: CS1. Read and interpret internal/external documents. CS2. Read and understand manuals, health and safety instructions, memos and other company documents. CS3. Read from different sources- books, screens in machines and signage. CS4. Interpret the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS5. Express statements or information clearly so that others can hear and understand. CS6. Participate in and understand the main points of simple discussions. CS7. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day to day activities

	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks.
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities. PS13. Ask relevant questions for better understanding.

UNIT 4 [This unit is about operation and management of a decision support system for precision agriculture]

Unit No.	04
Unit Title	Operation and Management of a Decision Support System for Precision Agriculture
Description	This unit profiles the functional knowledge, skills and competencies as well as job accountabilities needed for the jobholder to operate and manage a Decision Support System for Precision Agriculture.
Scope	This unit covers the following: • Decision support system set up • Acquisition of data on agricultural production • Data processing and analysis • Maintaining records of precision agriculture information
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Decision support system set up	To be competent, the individual must be able to: - PC1. Determine the requirements for data in form of data types, data attributes, data coordinate systems, datum, and map projections. - PC2. Set up a database management system for Precision agriculture in a Geographical Information System (GIS) environment.
Acquisition of data on agricultural production	To be competent, the individual must be able to: - PC3. Design a geo-referenced spatial sampling frames for data collection. - PC4. Collect geo-referenced data on agricultural production using data recorders. - PC5. Collect remotely-sensed data with Unmanned Aerial Vehicles (UAVs) such as drones.

Data processing and analysis	To be competent, the individual must be able to: PC6. Classify the data based on the digitizing and topology rules set forth during creation of database. PC7. Perform quality controls by checking for errors. PC8. Perform geo-processing through clipping, buffering, overlays etc. PC9. Create scripts and analysis models through flowcharts and processes. PC10. Perform statistical analyses and predictions to generate pertinent information for decision making. PC11. Prepare reports summarizing field productivity and profitability in graphical or tabular form.
Maintaining records of precision agriculture information	To be competent, the individual must be able to: PC12. Record information in written or electronic/magnetic form. PC13. Identify the underlying principles, reasons, or facts of information by breaking down information or data into separate parts.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Legislation, regulations, policies, standards, and procedures followed in the company relevant to own employment and performance conditions OK2. Organisational culture, vision and mission OK3. Company's service level agreements and policies OK4. Company's code of conduct
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Database creation and management TK2. Survey techniques and methods TK3. Geostatistical methods TK4. Statistical data analysis TK5. Spatial data analysis techniques TK6. Computer software programming TK7. Computational modelling and simulation
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Regulatory framework in the sector

A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions.
	Reading Skills
	The individual on the job must be able to: CS2. Read and interpret internal/external documents. CS3. Read and understand manuals, health and safety instructions, memos and other company documents. CS4. Read from different sources- books, screens in machines and signage. CS5. Interpret the various colour codes, nomenclature and acronyms related to the profession.
B. Professional Skills	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS6. Express statements or information clearly so that others can hear and understand. CS7. Participate in and understand the main points of simple discussions. CS8. Respond appropriately to any queries.
	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to:

	PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities. PS13. Ask relevant questions for better understanding.

UNIT 5 [This unit is about repair and maintenance of precision agriculture hardware and equipment]

Unit No.	05
Unit Title	Repair and Maintenance of Precision Agriculture Hardware and Equipment
Description	This unit profiles the functional knowledge, skills and competencies as well as job accountabilities needed for the job holder to repair and maintain Precision Agriculture technologies.
Scope	This unit covers the following: • Functional analysis of precision agriculture technology • Diagnostics and troubleshooting • Repair and maintenance
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Functional analysis of precision agriculture technology	To be competent, the individual must be able to: - PC1. Identify the components of precision agriculture technology. - PC2. Trace the functional processes of precision agriculture technology using appropriate equipment and tools.
Diagnostics and troubleshooting	To be competent, the individual must be able to: - PC3. Diagnose the malfunctioning of components of precision technology using appropriate equipment and tools. - PC4. Analyse the results of diagnosis to understand the nature of the faults. - PC5. Troubleshoot the faults using appropriate procedures, equipment and tools as contained in operators and service manuals. - PC6. Consult with manufacturers or suppliers of the hardware and equipment if the nature of the faults cannot be resolved with the available methods, procedures and tools.

Repair and maintenance	To be competent, the individual must be able to: PC7. Carry out servicing and maintenance activities in accordance with the operator's manual. PC8. Ensure regular servicing of the equipment and tools. PC9. Follow guidelines for storage of spare parts and other agriculture machineries/implements PC10. Maintain record of consumption of fuel/oil with reference to operating environment of the machines and equipment. PC11. Ensure the software and hardware system of the precision equipment are operating correctly. PC12. Ensure that repairs are carried out using genuine spare parts PC13. Ensure that the equipment and the precision accessories are calibrated properly according to the operations manual. PC14. Keep machines clean and check for leakage of oil, fuel, etc.
Knowledge and Understanding (K)	
A. Organisation-al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Legislation, regulations, policies, standards, and procedures followed in the company relevant to own employment and performance conditions OK2. Organisational culture, vision and mission OK3. Company's service level agreements and policies OK4. Company's code of conduct
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Mechatronics: mechanical and electronic systems integration TK2. Engineering principles of agricultural machines TK3. Agricultural engineering technology TK4. Agricultural machinery and equipment TK5. Basic mechanical systems TK6. Hydraulic and pneumatic systems TK7. Sensors and actuators TK8. Computers and logic systems TK9. Optimization of mechatronic systems TK10. Digital signal processing TK11. Software and data acquisition TK12. Measurement techniques: sensors and transducers

	TK13. Signal conditioning TK14. Computer-based instrumentation systems TK15. Data recording and logging
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Engineering Institution of Zambia rules and regulations RK2. Zambia Information Communications and Technology Authority (ZICTA) rules and regulations.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions.
	Reading Skills
	The individual on the job must be able to: CS2. Read and interpret internal/external documents. CS3. Read and understand manuals, health and safety instructions, memos and other company documents. CS4. Read from different sources- books, screens in machines and signage. CS5. Interpret the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS6. Express statements or information clearly so that others can hear and understand. CS7. Participate in and understand the main points of simple discussions. CS8. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow code of conduct.

	PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to:
	PS7. Recognize problems and search for solutions.
	PS8. Choose best methods to complete assigned tasks.
	PS9. Approach relevant authority when required.
	PS10. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to:
	PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to:
	PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities.
	PS13. Ask relevant questions for better understanding.

UNIT 6 [This unit is about occupational health, safety and environmental protection]

Unit No.	06
Unit Title	Occupational Health, Safety and Environmental Protection
Description	This unit profiles the functional knowledge, skills and competencies as well as job accountabilities needed for the job holder to promote health, safety and environmental protection in the work place.
Scope	This unit covers the following: • Health & safety regulations • Environmental protection
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Health & safety regulations	To be competent, the individual must be able to: - PC1. Read, interpret and implement national and organizational safety and health policies and regulations. - PC2. Assess risks and possible safety hazards of all aspects of operations.

Environmental protection.	To be competent, the individual must be able to: PC3. Read, interpret and implement the environmental policies for the organisation. PC4. Read, interpret and implement environmental standard operating procedures and policies of the organisation. PC5. Read, interpret and implement national and global environmental regulations.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: Company quality, health and safety policies OK1. Company environmental policies OK2. Company regulations and best practices
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Safety and health risk assessment TK2. Environmental risk assessment TK3. Toolbox talk
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Government regulatory agency requirements for health & safety
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions.
	Reading Skills
	The individual on the job must be able to: CS2. Read and interpret internal/external documents. CS3. Read and understand manuals, health and safety instructions, memos and other company documents. CS4. Read from different sources- books, screens in machines and signage. CS5. Interpret the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS6. Express statements or information clearly so that others can hear and understand. CS7. Participate in and understand the main points of simple discussions.

	CS8. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities. PS13. Ask relevant questions for better understanding.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

A Precision Agricultural Technologist need a range of equipment, tools and consumable materials including, but not limited to:

- Laptop computers or pads
- Internet
- Satellite or smart phones
- Differential GPS receivers
- Satellite imagery
- Aerial photographs
- Computer software applications
- Crop simulation models
- Data loggers
- Testing equipment and tools
- Tool box
- Personal protective equipment
- Handbooks
- Operators, service and manufacturers manuals

6. DILEMMAS/CHALLENGES AND COMPLEXITIES FOR A JOB HOLDER

A Precision Agricultural Technologist may face challenges such as:

- Differences in constructional and operational features of machines on the market.
- Obsolescence of equipment and tools due to rapid change of digital technology.
- Limited availability of tools, parts or spares locally.

7. ALTERNATIVE CHOICES (SOLUTIONS) TO DILEMMAS AND COMPLEXITIES

Solutions to the dilemmas and complexities include:

- Professional training with technology suppliers or manufacturers.
- Participating and technology fairs.
- Networking and sharing resources with other technology users.

8. WORKING CONDITIONS/ENVIRONMENT

A Precision Agricultural Technologist works with expensive and sensitive equipment and sensors, as well as diverse machines, equipment, tools, toxic substances and volatile materials. The Technologist is susceptible to fires, explosions, structural failures and equipment malfunctions. The working

conditions include cold, hot and wet weather, standing and walking for long hours, lifting materials, working in day or night shifts, noise and dust.

9. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE

9.1 Internal/Within the Organization

Parties involved/interacting with the Precision Agricultural Technologist who are internal to the organization include:

- Supervisors,
- Subordinates, and
- Other employees.

9.2 External/Outside the Organization

Parties involved/interacting with the Precision Agricultural Technologist who are external include:

- Customers/clients,
- Government regulators,
- Trainers,
- Suppliers of equipment/tools/consumables,
- Professional associations

10. PHYSICAL DEMANDS ON THE BODY

A Precision Agricultural Technologist demands:

- Physique to sustain strenuous conditions;
- Ability to walk and stand for long periods of time;
- Ability to bend, stretch, twist, or reach out;
- Ability to lift relatively heavy materials, tools and equipment;
- Ability to use fingers, hands and feet with ease to complete the assigned task (dexterity); etc

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

A.1.1 Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programmes developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programmes developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

A.1.2 Awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part and evaluations for skill practical part for their respective candidates.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.PAT.01		
ZQF Level	6	Version Number	01
Sector	Agriculture	Date of Approval	19 th May 2022
Sub Sector	Agricultural engineering	Date of Last Review	N/A
Occupation	Precision Agricultural Technologist	Date of Next Review	May 2027

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